

Final Experimental Report

Beat the Burn: The Effect of Music Tempo on Isometric Plank Endurance

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I. Introduction

This report studies the within-person causal effect of listening to high-tempo versus low-tempo instrumental EDM music on plank time-to-volitional-exhaustion (TTE), measured in seconds. The causal quantity of interest is the difference between each participant’s potential outcome under the high-tempo condition and their potential outcome under the low-tempo condition, averaged over the retained analytic sample. The primary estimand is therefore the average treatment effect inside the randomized crossover design: for each participant, we observe plank duration under each condition, take the within-person difference, and average it across participants.

An observational comparison would not identify that estimand. People who naturally prefer fast music may also differ systematically in baseline strength, training history, motivation, pain tolerance, or willingness to push through discomfort. Those selection effects would contaminate any simple comparison of high-tempo listeners to low-tempo listeners. The experiment was therefore designed to randomize treatment order and let each participant serve as their own control, sharply reducing stable between-person confounding.

The empirical literature on music and isometric endurance is mixed, which makes the specific question of tempo – holding genre, lyrics, and volume constant – worth testing under a clean within-subjects design. Poon et al. (2024), using a randomized crossover trial, found that music during the bout extended time-to-exhaustion in both plank and wall-sit tasks and delayed fatigue perception without detectable differences in heart rate or blood lactate, a pattern consistent with a psychophysical rather than metabolic pathway. Feiss et al. (2021), in a between-subjects design, reported that fast- and slow-tempo music prolonged dissociative attention during a wall-sit but not during a plank, and found no tempo-driven improvement in plank endurance – a null result likely shaped by the large between-person variance that a between-subjects design cannot absorb. The Terry et al. (2020) meta-analysis of 139 studies reported a generalized music effect of Hedges’ $g = 0.31$ on physical performance, while Centala et al. (2020) found that fast-tempo music raised the EMG-based fatigue threshold and maximum power output without increasing end-exercise perceived exertion, consistent with improved tolerance to discomfort. Ballmann (2021) further emphasizes that preference and hedonic response moderate effect sizes, which motivated our use of genre-consistent instrumental EDM tracks on both arms. Two mechanisms are plausible for isometric tasks specifically: psychological dissociation, in which fast external stimulation competes with interoceptive fatigue cues and lowers perceived exertion, and physiological arousal, in which sympathetic activation raises motivation and tolerance for discomfort near task failure.

Our pre-registered primary hypothesis was directional: high-tempo music would increase plank time-to-volitional-exhaustion relative to low-tempo music. Two secondary hypotheses follow from the mechanisms above. Under the arousal pathway, the high-tempo condition should produce higher self-reported motivation

and perceived energy. Under the dissociation pathway, the high-tempo condition should not produce proportionally higher perceived exertion despite longer holds, consistent with music competing with fatigue signals rather than simply inducing participants to try harder. These secondary outcomes are therefore tests of mechanism rather than of overall performance.

II. Experimental Details

The treatment condition was defined as high-tempo instrumental EDM at or above 125 BPM, while the control condition was low-tempo instrumental EDM at or below 80 BPM. Participants received fixed listening links rather than choosing their own songs, were instructed to use headphones, and were asked to keep playback near a common reference volume of roughly 60 percent. The participant-facing app kept the labels neutral during the sessions, while the analysis pipeline mapped the underlying tracks into the high-tempo and low-tempo treatment definitions. This standardization kept the treatment as consistent as possible across participants in a remote setting.

Audio stimuli. The study used two fixed YouTube audio sources, one assigned to the high-tempo arm and one to the low-tempo arm. The high-tempo stimulus was *Best Uplifting Trance Mix - All Time Favourites #1* by The Grand Sound, streamed from <https://youtu.be/iXAbte4QXKs?si=3iLV6OhMxesZBYtZ>. The low-tempo stimulus was *Easy Pop Backing Track Without Bass* by Gonella, MusicArt Rapallo, streamed from <https://youtu.be/x8WbHWwZixk?si=HnDa6uBvIK6j-kcc>. During data collection, the app presented these tracks under neutral session labels (Audio A and Audio B) so tempo was not revealed in the participant interface; the preprocessing pipeline then mapped the stored track identifiers to the `high_tempo` and `low_tempo` analysis labels used throughout this report.

Data acquisition was conducted through a custom mobile-first web app rather than a survey-only workflow. Participants registered through the app, received direct session links, completed guided plank sessions with embedded audio and pre/post task questions, and had their responses written through a Google Apps Script backend into Google Sheets. The app was also built with bilingual English and Spanish participant flows, which made the study more accessible and helped include participants across different locales and countries rather than limiting the interface to a single language. The participant roster stored non-PII baseline metadata such as age, gender, locale, and planned session timing, while the session records captured assigned audio condition, plank duration, and survey responses. The app also offered an optional image-capture feature that, for participants who enabled it, generated a contact-sheet style image stream that was used for plank-form quality control and to help validate timing. The preprocessing pipeline then merged those records into a participant-level wide file and the canonical derived datasets used by the analysis reports.

The pilot and design logic pointed in two directions at once. First, natural baseline variance in plank endurance is large enough that a between-subjects design would be noisy and inefficient. Second, repeated isometric effort creates obvious risks of fatigue, learning, and carryover across sessions. The final experiment therefore used a randomized crossover design in which Session 2 was scheduled within a 24 to 72 hour window after Session 1, while the final crossover analysis excluded cases where the realized elapsed time between recorded sessions was under 24 hours. That choice was intended to preserve the efficiency benefits of within-person comparison while reducing temporal spillovers from the first plank attempt into the second.

Ex Ante Power Analysis

Before data collection, we conducted a simulation-based power analysis grounded in prior literature and calibrated with pilot data. The assumed baseline plank hold time was 111 seconds with a population standard deviation of 43 seconds, drawn from Feiss et al. (2021) and consistent with the 40.9 to 58.3 second range reported in Poon et al. (2024). We simulated three literature-based effect-size scenarios spanning the range of prior findings: a conservative +5 second effect reflecting the near-null results in Feiss et al. (2021) and Centala et al. (2020), a moderate +13 second effect corresponding to the generalized Hedges' $g = 0.31$ reported in the Terry et al. (2020) meta-analysis of 139 studies, and an optimistic +26 second effect drawn from the Poon et al. (2024) crossover trial (Cohen's $d = 0.76$). We then ran a pilot-calibrated simulation using our own eight-participant pilot, which produced a within-person paired effect of +15.6 seconds and a

paired-difference standard deviation of 27.65 seconds. Under the pilot-calibrated crossover simulation the within-subjects design was anticipated to reach 80 percent power at approximately 20 participants and to exceed 93 percent power at 40 participants, whereas a parallel between-subjects design would have required roughly 260 participants to reach 80 percent power under the optimistic scenario. We targeted a sample of approximately 46 participants, a size at which the crossover design was expected to deliver greater than 90 percent power if the true effect matched the pilot or Poon et al. benchmarks, while remaining informative under the Terry et al. meta-analytic scenario.

Randomization and Balance

Treatment-order assignment was implemented through the registration app, which randomized each eligible participant to one of the two crossover sequences (high-tempo first or low-tempo first). Balance on observed baseline characteristics was not perfect. The high-tempo-first arm was on average approximately nine years younger than the low-tempo-first arm (mean age 31.2 versus 40.3 years; see Table 20 in Section VI), a meaningful imbalance in a sample of 46 participants. We treat this as a randomization deviation to flag rather than as a source of primary bias for three reasons. First, the primary crossover estimator compares each participant to themselves, so stable participant-level characteristics like age do not enter the within-person contrast and therefore cannot bias the treatment estimate on that pathway. Second, the first-period benchmark in Section VI (Model B2) includes age as a baseline covariate, which addresses the same imbalance directly in the weaker between-person design. Third, the heterogeneity analysis in Section VII (Models H1 through H4) finds no evidence that the treatment effect varies with age after Bonferroni correction, which is the nearest substantive test of whether the imbalance materially affects the conclusions. The imbalance is therefore flagged and accommodated rather than left as an unaddressed threat.

Model Roadmap

Table 1: Model roadmap for the final report

family	id	model	sample	purpose
P: Primary crossover	P1	Raw paired contrast	38 completed pairs	Direct within-person treatment contrast
P: Primary crossover	P2	Fixed-effects without period	76 repeated-measures rows	Linear-model bridge without period adjustment
P: Primary crossover	P3	Fixed-effects with period	76 repeated-measures rows	Main regression specification with period control
D: Diagnostics	D1	Broad period comparison	76 repeated-measures rows	Checks broad Session 2 drift
D: Diagnostics	D2	Sequence diagnostic regression	76 repeated-measures rows	Checks whether period patterns differ by randomized order
B: Benchmark	B1	First-period raw contrast	46 Session 1 observations	Emulates a parallel-arm comparison using Session 1 only
B: Benchmark	B2	First-period adjusted regression	46 Session 1 observations	Adds baseline adjustment to the Session 1 benchmark
S: Secondary outcomes	S1-S3	Paired secondary contrasts	38 completed pairs	Tests simple paired shifts in RPE, motivation, and energy
S: Secondary outcomes	S4-S6	Fixed-effects secondary models	76 repeated-measures rows	Re-estimates secondary outcomes in the FE framework

family	id	model	sample	purpose
H: Heterogeneity	H1- H4	Treatment-by- subgroup interaction models	76 repeated-measures rows	Tests whether treatment effects differ by pre-specified subgroup
R: Randomiza- tion inference	R1	Bridge linear model	38 participant-level differences	Writes the paired estimand in regression form
R: Randomiza- tion inference	R2	Randomization inference	38 participant-level differences	Provides design-based sign-flip p-values under the sharp null

III. Participant Flow and Attrition

The analysis begins with the full registered sample and only then narrows to the retained analytic populations. That ordering keeps attrition, missingness, and protocol exclusions visible before any treatment effects are interpreted.

Reading the flow correctly

This step intentionally starts from the **full initial distribution** and only then narrows to the analytic samples.

- First, we show the full registered roster and how many participants reach each observation stage.
- Next, we separate **attrition** (for example, not returning for session 2) from **compliance / protocol exclusions**.
- Only after those counts are visible do we define the **primary analytic sample** and the **first-period benchmark sample**.

This mirrors the report plan: the reader should see the whole study population first, then exactly why some participants are not used for a given estimand.

Build participant flow dataset

The participant roster is the base denominator for flow accounting. We left-join the experiment wide file onto the full participant list so registered participants with zero observed sessions still appear in the flow dataset.

Participant flow summaries

Table 2: Table A. Participant flow

population	n
Registered participants	58
Randomized order assigned	58
Session 1 observed	47
Session 2 observed	40
Completed both sessions	40
Primary analytic sample	38
First-period benchmark sample	46

Participant flow from registration to analytic samples
Full distribution first, then attrition and protocol-based exclusions

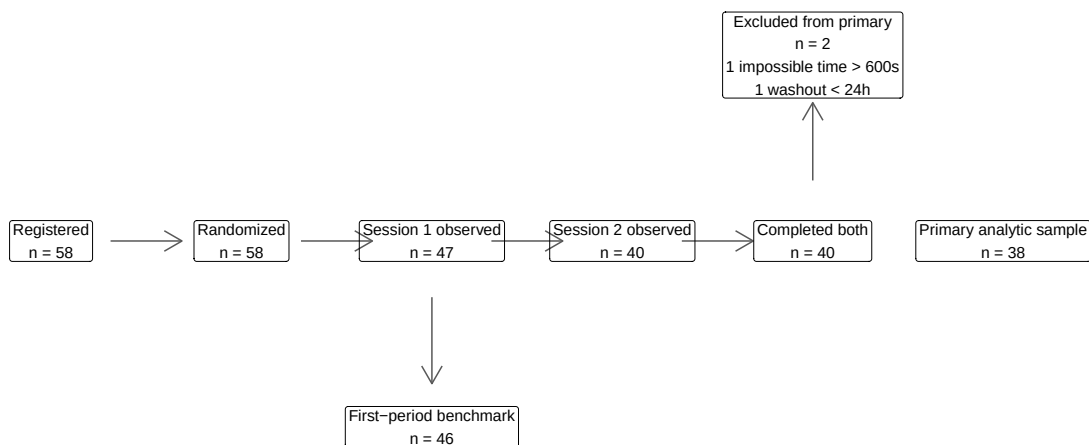


Table 3: Table B. Attrition by randomized order

assigned_order	started_session1	dropped_before_session2	retained_completed_both
high_tempo_first	23	4	19
low_tempo_first	24	3	21

Attrition-bias check

This check asks whether retention through both observed sessions differed meaningfully by randomized order arm or baseline demographics among randomized participants. It is intentionally light-touch: the goal is to surface obvious selection patterns without overfitting a small attrition model.

Table 4: Table C. Baseline demographics by retention status

retained_group	n	mean_age	sd_age	share_male
Completed both sessions	40	36.30	13.15	70.0
Did not complete both	18	32.67	10.05	66.7

Table 5: Table D. Retention by randomized order

assigned_order	randomized_participants	retained_completed_both	not_retained	retention_rate_pct
high_tempo_first	30	19	11	63.3
low_tempo_first	28	21	7	75.0

Table 6: Table E. Baseline gender counts by retention status

retained_group	male_n	female_n	male_share_pct
Completed both sessions	28	12	70.0
Did not complete both	12	6	66.7

Table 7: Table F. Attrition-bias diagnostic p-values

comparison	p_value
Retention by randomized order	0.4022
Baseline age by retention status	0.2556
Baseline gender by retention status	0.7996

These diagnostics should be read as descriptive checks rather than proof that attrition was ignorable. They focus on retention through Session 2 among randomized participants and cannot rule out selection on unobserved factors. They are also conceptually separate from the protocol-based primary-analysis exclusions, which apply only after a participant completed both observed sessions.

Table 8: Table G. Compliance and exclusion reasons

issue	n
Impossible time > 600s	1
Washout violation < 24h	1
Missing one condition	18
Excluded from primary analysis	2

Taken together, these outputs show a clear progression from 58 registered participants to 40 completed-both observations and finally to 38 protocol-valid pairs. In Table G, the 20 non-compliers were participants who either registered but did not complete any study session or completed Session 1 but did not return for Session 2. In this study, therefore, non-compliance primarily reflects attrition within the two-session crossover protocol rather than refusal of the assigned condition itself. We did not systematically collect formal reasons for attrition, so any explanation remains descriptive rather than definitive. Based on participant communications and team observation, the most plausible reasons were failure to allocate time to complete both sessions and discomfort during or after a session that discouraged continued participation. For example, one participant reported abdominal discomfort related to a prior surgery after Session 1 and was therefore unable to complete Session 2. The attrition-bias tables further suggest that retention through both sessions was not strongly associated with randomized order or the observed baseline demographics reported there, although those checks remain descriptive rather than definitive.

IV. Primary Crossover Estimate

The core estimate comes from the crossover sample of 38 protocol-valid completed pairs. In raw paired terms, mean TTE increased by 14.37 seconds under high tempo, which corresponds to a 16.3 percent improvement relative to the low-tempo mean. To make the specification ladder transparent, the report shows the raw paired contrast, then a participant fixed-effects model without period adjustment, and finally the period-adjusted fixed-effects crossover regression with participant-clustered standard errors.

Analytic sample

Table 9: Primary crossover sample

quantity	value
Paired analytic participants	38
Repeated-measures rows used in FE model	76
High-tempo first	19
Low-tempo first	19

quantity	value
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Descriptive summaries

Table 10: Paired descriptive summaries

quantity	value
Mean high-tempo TTE	102.440
Mean low-tempo TTE	88.071
Mean paired difference (high - low)	14.369
SD paired difference	28.921
Median paired difference	8.708
IQR paired difference	25.199
Independent participants	38.000

Compact metrics summary

Table 11: Requested summary metrics

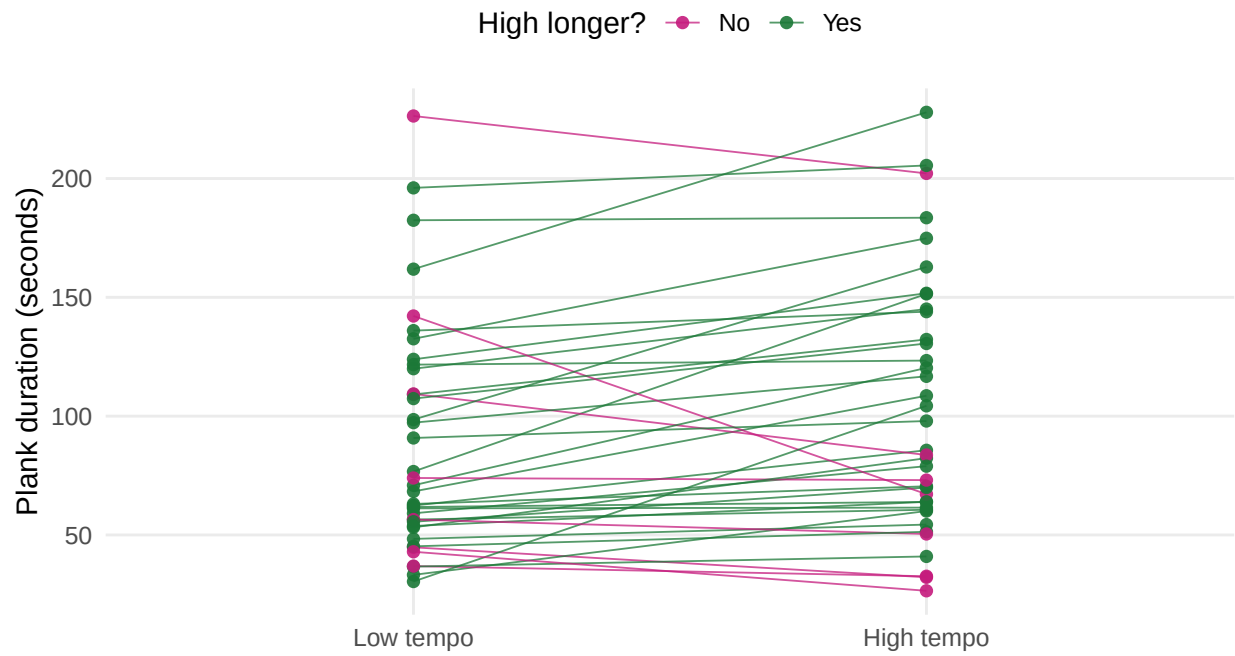
metric	value
Mean treatment effect (high - low, seconds)	14.37
Percentage improvement vs low tempo mean	16.3%
Standard deviation of the paired difference	28.92
Paired t-test p-value	0.0041
Completion rate (completed both / registered)	69.0%

Exploratory figures

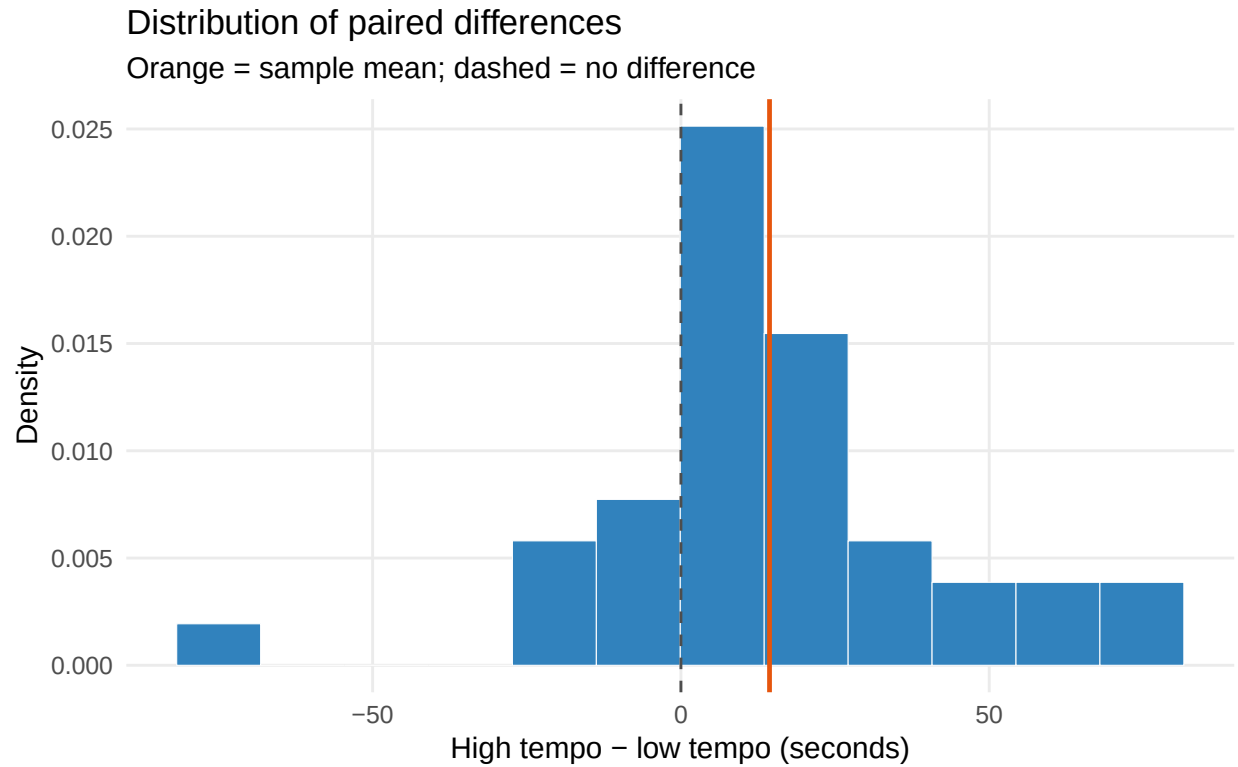
Spaghetti plot

Paired plank times by music condition

n = 38 primary-sample participants



Paired-difference distribution



Model 1: raw paired contrast

Table 12: Model 1. Raw paired contrast

quantity	value
Mean high-tempo TTE	102.4404
Mean low-tempo TTE	88.0709
Mean effect (high - low)	14.3694
95% CI lower	4.8633
95% CI upper	23.8756
Paired t statistic	3.0628
Paired p-value	0.0041
Standard error	4.6916
Paired effect size dz	0.4968

Model 2: fixed-effects crossover regression without period adjustment

Table 13: Model 2. Fixed-effects crossover regression without period adjustment

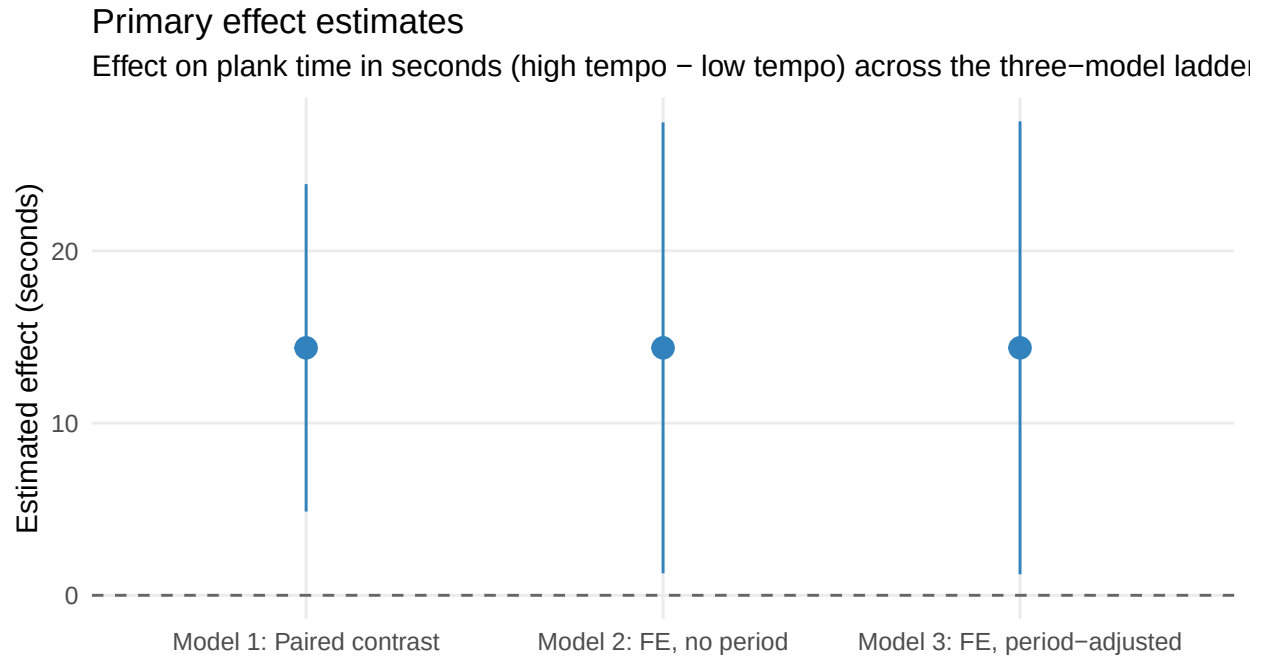
term	estimate	std_error	conf_low	conf_high	p_value
High-tempo indicator	14.3694	6.6796	1.2774	27.4615	0.0381

Model 3: fixed-effects crossover regression with period adjustment

Table 14: Model 3. Fixed-effects crossover regression with clustered SEs

term	estimate	std_error	conf_low	conf_high	p_value
High-tempo indicator	14.3694	6.7101	1.2177	27.5212	0.0391
Period 2 indicator	3.8441	6.7101	-9.3076	16.9958	0.5703

Main-effect comparison figure



The **stargazer** summary below isolates the regression members of the primary family, P2 and P3, so the reader can compare the fixed-effects specifications directly.

	P2: FE no period	P3: FE with period
High-tempo indicator	14.369** (6.680)	14.369** (6.710)
Period 2 indicator		3.844 (6.710)
Observations	76	76
R ²	0.920	0.922
Adjusted R ²	0.838	0.837

Note: *p<0.1; **p<0.05; ***p<0.01
This table covers the regression members of the primary family; P1 is a paired t-test and is reported outside stargazer.
Both models use the repeated-measures crossover sample.
Standard errors are clustered at the participant level.

The graphical story is consistent with the regression story: most of the within-person signal leans toward longer holds under high tempo, and the center of the paired-difference distribution lies to the right of zero. The three-model ladder then shows that the estimated treatment effect remains near 14.37 seconds across specifications, which is exactly the pattern one would hope to see if the result is design-driven rather than specification-driven.

V. Period and Order Diagnostics

Because the protocol required two plank attempts per participant, it is important to check whether broad session-order effects disrupted interpretation. The diagnostics below test whether performance shifted systematically from Session 1 to Session 2 and whether those patterns differed by randomized sequence arm.

Period summaries

Table 15: Session 1 vs Session 2 summaries

period	n	mean_tte	sd_tte	median_tte	iqr_tte
1	38	93.334	46.730	84.646	68.303
2	38	97.178	55.207	70.692	75.268

Period-by-order summaries

Table 16: Period-by-order diagnostic summary

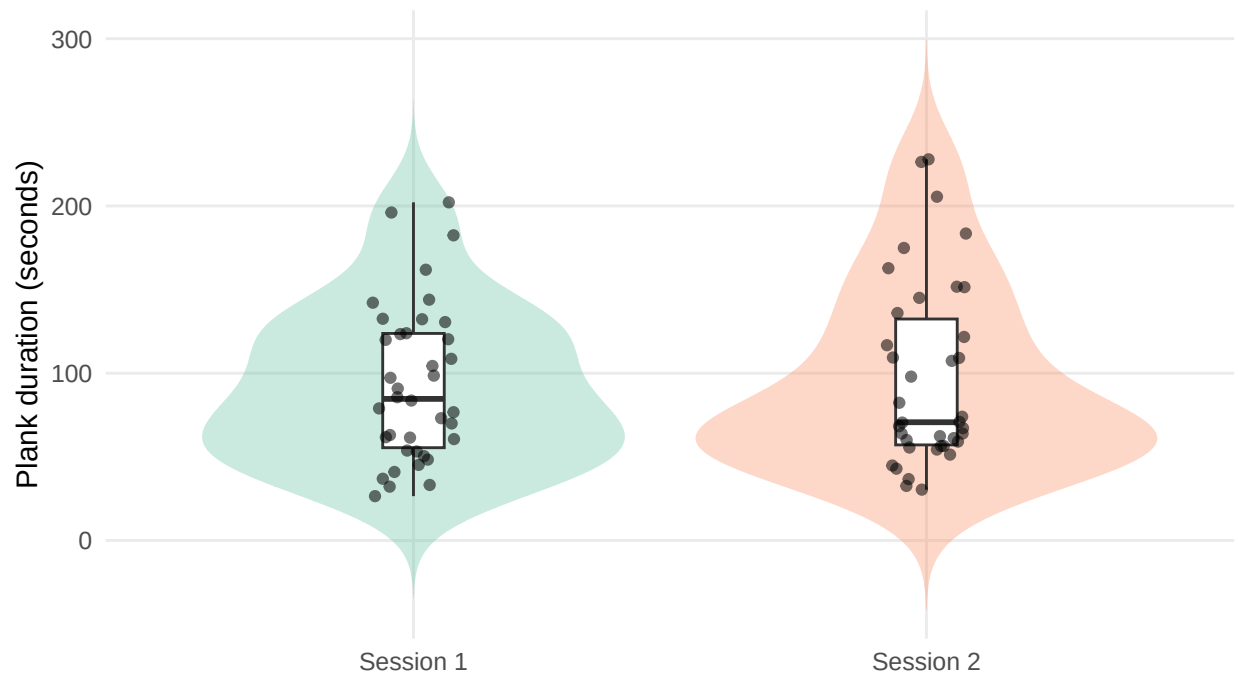
crossover_order	period	treatment	n	mean_tte	sd_tte	mean_rpe
high_tempo_first	1	high_tempo	19	91.001	44.336	6.895
high_tempo_first	2	low_tempo	19	80.476	46.395	6.263
low_tempo_first	1	low_tempo	19	95.666	50.116	7.105
low_tempo_first	2	high_tempo	19	113.880	59.359	7.632

Session-period figures

Session 1 vs Session 2 boxplot

Outcome distributions by session period

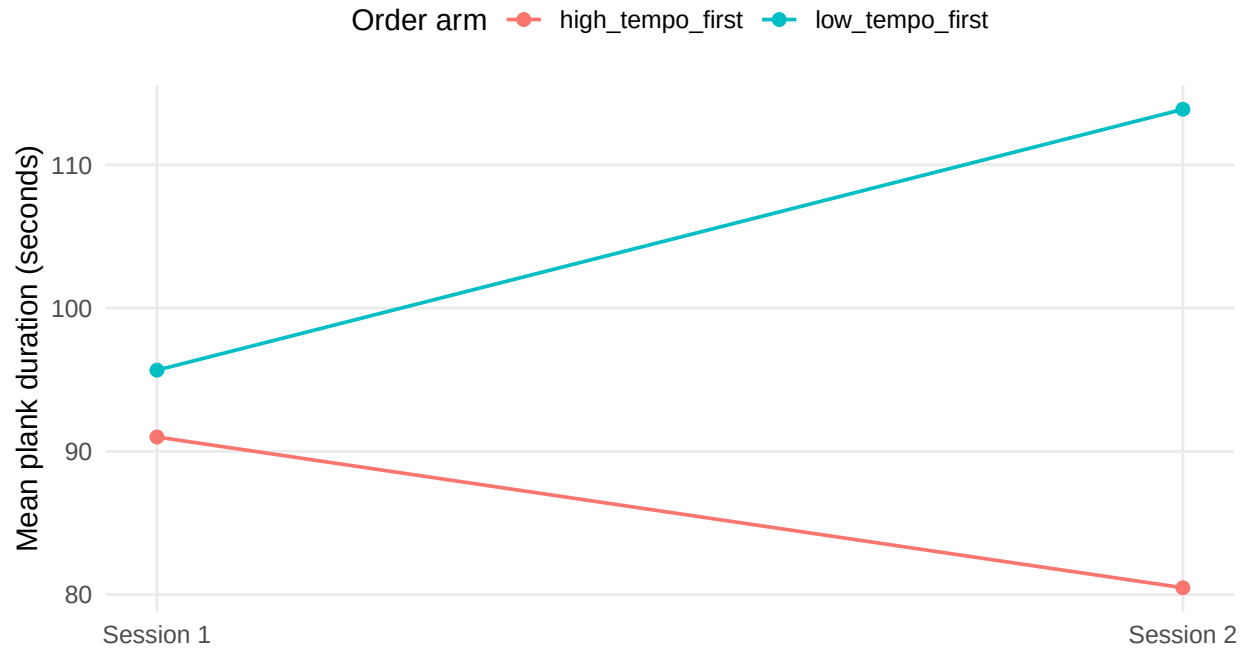
Primary analytic sample, n = 38



Order-arm trajectories across periods

Average plank time by order arm across periods

Checks for sequence or learning patterns



Diagnostic models

Broad period check

Table 17: Broad period comparison

quantity	value
Mean Session 2	97.1777
Mean Session 1	93.3336
Difference (Session 2 - Session 1)	3.8441
95% CI lower	-19.5456
95% CI upper	27.2339
p-value	0.7441

Sequence diagnostic regression

Table 18: Sequence diagnostic regression

term	estimate	std_error	p_value
period_factor2	-10.5253	5.8264	0.0750
crossover_orderlow_tempo_first	4.6650	15.4541	0.7636
period_factor2:crossover_orderlow_tempo_first	28.7389	9.4895	0.0034

Interpretation notes

The next **stargazer** output focuses on D2, the sequence diagnostic regression. It is included separately because the interaction term is the main formal check for order-specific period patterns.

	Plank time to exhaustion (seconds)
Period 2 indicator	−10.525* (5.826)
Low-tempo-first order	4.665 (15.454)
Period 2 x low-tempo-first	28.739*** (9.489)
Observations	76
R ²	0.057
Adjusted R ²	0.018

Note: *p<0.1; **p<0.05; ***p<0.01
This table covers the regression member of the diagnostic family; D1 is the broad period t-test and is reported outside stargazer.
The interaction term is the key validity check.
Standard errors are clustered at the participant level.

These diagnostics tell two stories. First, the broad Session 2 minus Session 1 difference pooled across arms averages about 3.84 seconds, which is small relative to the treatment effect. Second, however, the sequence diagnostic regression reveals a large and statistically significant period-by-order interaction. In a crossover design, this is the classical signature of either asymmetric carryover – in which completing one condition alters the response under the other – or of differential practice and fatigue between arms that is unrelated to treatment spillover. We cannot fully adjudicate between these explanations within the primary design alone. Two observations nonetheless bound the severity of the threat. First, the treatment effect points in the hypothesized direction within both randomized arms; the interaction reflects differing magnitudes of effect rather than a sign flip. Second, the first-period benchmark reported in Section VI is mechanically carryover-free by construction, because it uses only each participant’s Session 1 observation and therefore cannot be contaminated by any order effect. That benchmark produces a positive point estimate in the same direction as the primary crossover result. Its standard error is much larger, as the precision comparison makes clear, but its directional agreement with the primary estimate is evidence that the main treatment signal is not purely an artifact of ordering.

VI. First-Period Benchmark and Precision

The first-period benchmark asks what the treatment effect would have looked like if the study had been analyzed as a conventional parallel-arm trial using one independent observation per participant. That benchmark is supplementary by design and serves two purposes. Its primary purpose is to show how much precision the crossover design gained by conditioning on stable participant-level differences rather than trying to estimate the effect entirely across people. Its secondary purpose is to provide a carryover-free sensitivity check on the primary estimate: because the benchmark uses only each participant’s Session 1 observation, it is mechanically immune to the period-by-order interaction flagged in Section V.

The contrast is stark in this dataset: the crossover standard error is 4.69 seconds, whereas the first-period benchmark standard error is 13.87 seconds, a ratio of about 2.96 to 1. The benchmark estimate therefore illustrates the design argument as much as the substantive treatment argument.

Benchmark sample summary

Table 19: First-period benchmark sample

quantity	value
Benchmark participants	46
Assigned high tempo first	23
Assigned low tempo first	23

quantity	value
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Balance checks

Table 20: Compact randomization check artifact: counts, age, gender, and baseline activity

treatment_label	n	mean_age	share_male	share_active_at_least_weekly
High tempo first	23	31.17	73.9	34.8
Low tempo first	23	40.35	60.9	47.8

Table 21: Balance check: age by assigned first condition

treatment_label	n	mean_age
High tempo first	23	31.174
Low tempo first	23	40.348

Table 22: Balance check: gender counts

treatment_label	female	male
High tempo first	6	17
Low tempo first	9	14

Table 23: Balance check: baseline activity counts

treatment_label	1	2	3	4
High tempo first	9	6	1	7
Low tempo first	7	5	0	11

The first-period benchmark arms are balanced in **counts** but not perfectly identical on observables. The main age difference is visible here, which is worth acknowledging. It is less threatening for the primary estimate because the main analysis uses the crossover design and participant fixed effects, which absorb stable participant-level differences.

Benchmark descriptives

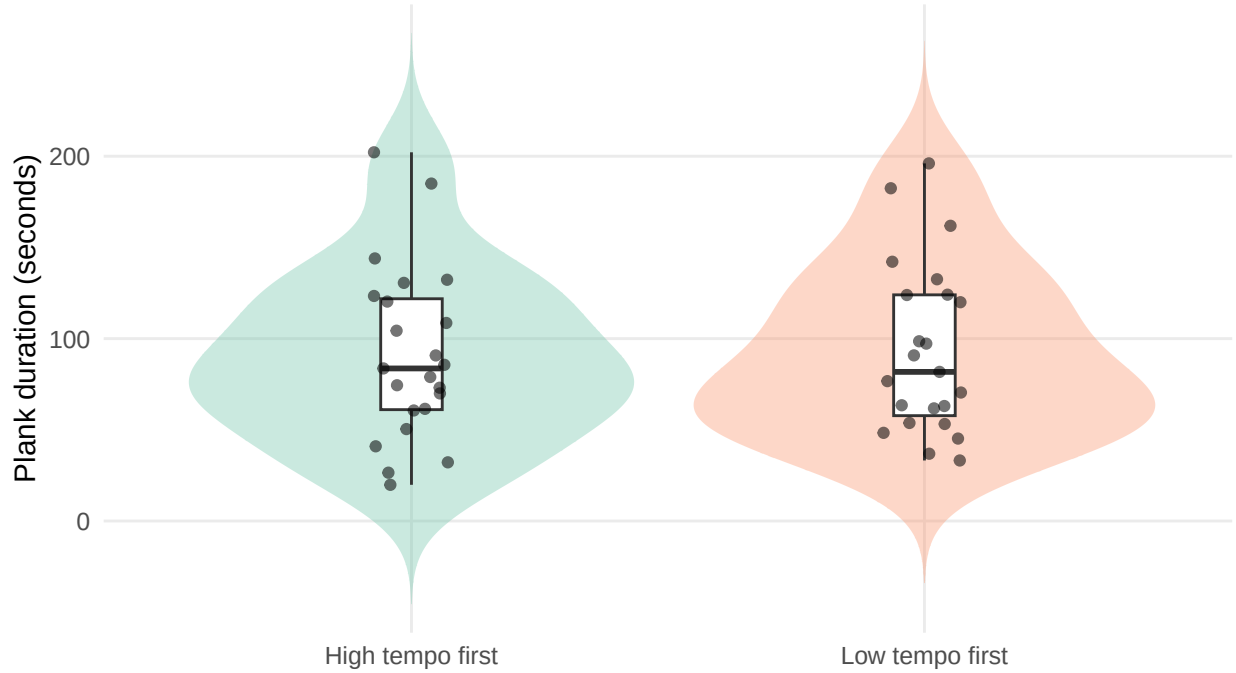
Table 24: Session 1 descriptives by assigned condition

treatment_label	n	mean_tte	sd_tte	median_tte	iqr_tte
High tempo first	23	91.265	47.460	83.655	60.795
Low tempo first	23	93.801	46.614	81.760	66.236

Figure

Session 1 benchmark by assigned first condition

n = 46 independent participants



Benchmark Model 1: raw contrast

Table 25: Benchmark Model 1. Raw period-1 contrast

quantity	value
Mean high-first session 1 TTE	91.2654
Mean low-first session 1 TTE	93.8013
Difference (high first - low first)	-2.5359
95% CI lower	-25.4194
95% CI upper	30.4912
t statistic	0.1828
p-value	0.8558

Benchmark Model 2: precision-adjusted regression

Table 26: Benchmark Model 2. Precision-adjusted regression

term	estimate	std_error	p_value
High-tempo first	-6.6796	15.0537	0.6595
Participant age	-0.1142	0.6001	0.8499

The benchmark `stargazer` table below summarizes B1 and B2, the two Session 1-only models. These are useful as a design contrast because they show how weak the treatment evidence becomes once the analysis stops comparing participants to themselves.

		Plank time to exhaustion (seconds)	
	B1: Raw contrast		B2: Adjusted benchmark
High-tempo first	−2.536 (13.871)		−6.680 (15.054)
Participant age			−0.114 (0.600)
Male participant			23.733 (14.943)
Observations	46		46
R ²	0.001		0.057
Adjusted R ²	−0.022		−0.010

Note: *p<0.1; **p<0.05; ***p<0.01
Both models use Session 1 only to emulate a parallel-arm design.
B2 adds baseline adjustment to the raw first-period contrast.
Standard errors are classical OLS because each participant contributes one observation.

Precision calculations

Precision table

Table 27: Realized precision by design

design	independent_n	observations	relevant_sd	standard_error	ci_width_95	mde_80_power
Primary crossover	38	76	28.9211	4.6916	18.3908	13.1440
First-period benchmark	46	46	47.0386	13.8709	54.3730	38.8606

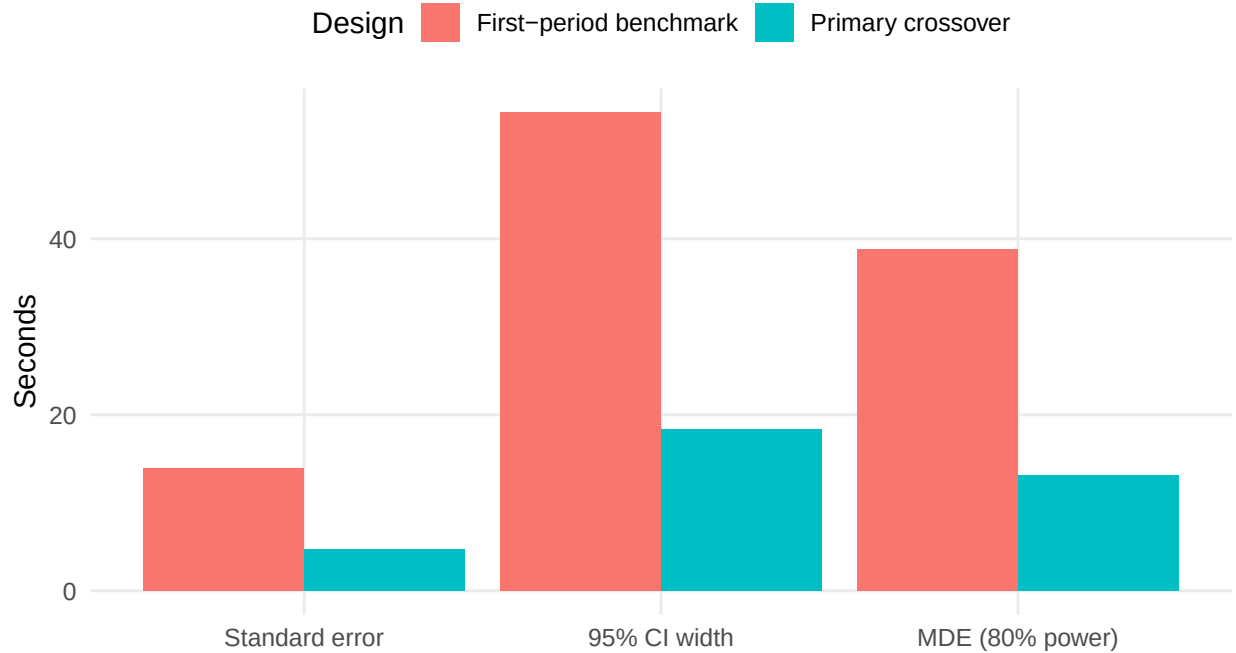
Table 28: Parallel-design sample size needed to match crossover precision

quantity	value
Observed benchmark total N	46
Required benchmark total N to match crossover SE	404
Required benchmark N per arm to match crossover SE	202
Extra independent participants needed	358

Precision comparison figure

Realized precision comparison

Smaller values indicate tighter design precision



Interpretation notes

Table 29: Relative precision loss in the benchmark design

quantity	value
Parallel SE / crossover SE	2.9565
Parallel CI width / crossover CI width	2.9565

The key takeaway is not merely that the benchmark p-value is larger; it is that the entire benchmark design is far less informative. The benchmark treatment estimate is noisy and non-significant, while the precision tables show that a parallel-arm design would have needed far more independent participants to reach the same level of precision achieved by the crossover analysis.

VII. Exploratory Mechanisms and Heterogeneity

The secondary analyses ask why the treatment may have worked and whether it appears to work differently across subgroups. The mechanism checks focus on RPE, motivation, and energy; the heterogeneity checks examine age, gender, baseline activity level, and plank familiarity. Because subgroup exploration creates a multiple-comparisons problem, the report applies a Bonferroni correction and treats the HTE results as exploratory.

In directional terms, motivation shows the clearest secondary signal (0.474, $p = 0.0371$), while the adjusted subgroup p-values remain uniformly unconvincing (1.000, 1.000, 1.000, 1.000).

Secondary outcomes: paired contrasts

Table 30: Exploratory paired contrasts for secondary outcomes

outcome	mean_high	mean_low	mean_diff_high_minus_low	conf_low	conf_high	p_value
RPE	7.2632	6.6842	0.5789	-0.0161	1.1740	0.0562
Motivation	5.1842	4.7105	0.4737	0.0298	0.9176	0.0371
Energy	4.6053	4.3158	0.2895	-0.1748	0.7537	0.2144

Secondary outcomes: fixed-effects versions

Table 31: Exploratory fixed-effects treatment coefficients for secondary outcomes

outcome	estimate	std_error	p_value
RPE	0.5789	0.4237	0.1803
Motivation	0.4737	0.2905	0.1117
Energy	0.2895	0.3259	0.3803

HTE setup

We estimate treatment interactions on the primary TTE outcome. The coefficient of interest in each model is the **treatment-by-subgroup interaction**, which asks whether the high-tempo effect changes across participants with different characteristics.

Because subgroup results are exploratory, we report Bonferroni-adjusted p-values across the four planned subgroup tests.

The plank-familiarity proxy comes from the question “**Do you regularly perform planks?**”. To avoid treating two slightly different “occasionally” codes as distinct levels, we collapse the raw 4-level coding into a cleaner 3-step ordinal variable:

- 1 = rarely or never
- 2 = occasionally
- 3 = regularly

HTE models

Table 32: Exploratory HTE interaction results with Bonferroni correction

Subgroup	Interaction	Estimate	SE	p	Bonferroni p
Age	High tempo x age	0.2723	0.5090	0.5960	1
Gender	High tempo x gender	5.5517	13.5212	0.6839	1
Activity level	High tempo x activity	-0.8812	5.3176	0.8693	1
Plank familiarity	High tempo x familiarity	2.8831	9.7486	0.7692	1

This **stargazer** block summarizes the fixed-effects secondary-outcome models, **S4–S6**. It complements the paired secondary tables by showing whether the same directional patterns remain when the outcomes are re-estimated in the crossover regression framework.

	RPE S4: RPE FE	Motivation S5: Motivation FE	Energy S6: Energy FE
High-tempo indicator	0.579 (0.424)	0.474 (0.290)	0.289 (0.326)
Period 2 indicator	-0.053 (0.424)	0.526* (0.290)	0.237 (0.326)
Observations	76	76	76
R ²	0.788	0.762	0.606
Adjusted R ²	0.559	0.505	0.178
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01 These are the regression members of the secondary-outcome family; S1-S3 are paired t-tests reported outside stargazer. All three models use participant fixed effects and a Period 2 control. Standard errors are clustered at the participant level.			

The final exploratory **stargazer** block summarizes the heterogeneity family, H1-H4. Here the interaction term is the key quantity because it tests whether the treatment effect varies across the pre-specified subgroup dimension in each column.

	H1: Age	H2: Gender	H3: Activity	H4: Familiarity
Plank time to exhaustion (seconds)				
High-tempo main effect	14.369** (6.757)	10.425 (8.769)	16.712 (13.984)	10.197 (17.301)
Moderator main effect	-0.433* (0.254)	25.506*** (6.761)	-4.451 (2.659)	-11.225** (4.874)
Period 2 indicator	2.475 (6.611)	4.282 (7.355)	4.006 (6.514)	4.527 (6.895)
High tempo x moderator	0.272 (0.509)	5.552 (13.521)	-0.881 (5.318)	2.883 (9.749)
Observations	76	76	76	76
R ²	0.923	0.922	0.922	0.922
Adjusted R ²	0.834	0.833	0.832	0.833
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01 The coefficient of interest changes across columns: the treatment-by-subgroup interaction is the HTE test. All models use participant fixed effects and a Period 2 control. Standard errors are clustered at the participant level; multiplicity adjustment is discussed in the main report.				

These outputs suggest that the treatment may have worked partly through engagement or readiness to perform, because motivation moves in the expected direction more clearly than the other secondary outcomes. At the same time, the heterogeneity table does not support strong claims that the treatment was materially larger or smaller for any of the tested subgroups after Bonferroni correction.

VIII. Design-Based Inference Check

The final inferential step asks whether the observed paired treatment effect looks unusual under the sharp null implied by the randomization mechanism itself. The bridge linear model writes the observed paired difference in regression form, while the randomization-inference procedure uses 10,000 random sign flips of the participant-level treatment differences to generate a distribution-free p-value.

Observed statistic

Table 33: Observed statistic for randomization inference

quantity	value
Participants	38.0000
Observed paired mean difference (high - low)	14.3694

Bridge linear model

The randomization-inference statistic is the same paired mean difference reported in the primary analysis. To make that link explicit, we first write the observed effect as the intercept from a one-sample linear model on the participant-level differences. This does not change the estimand; it only provides the conventional regression-style coefficient, standard error, and p-value before the design-based RI calculation.

Table 34: Bridge linear model for the paired estimand

quantity	value
Intercept (mean paired effect)	14.3694
Standard error	4.6916
t statistic	3.0628
Conventional two-sided p-value	0.0041
95% CI lower	4.8633
95% CI upper	23.8756

Randomization distribution

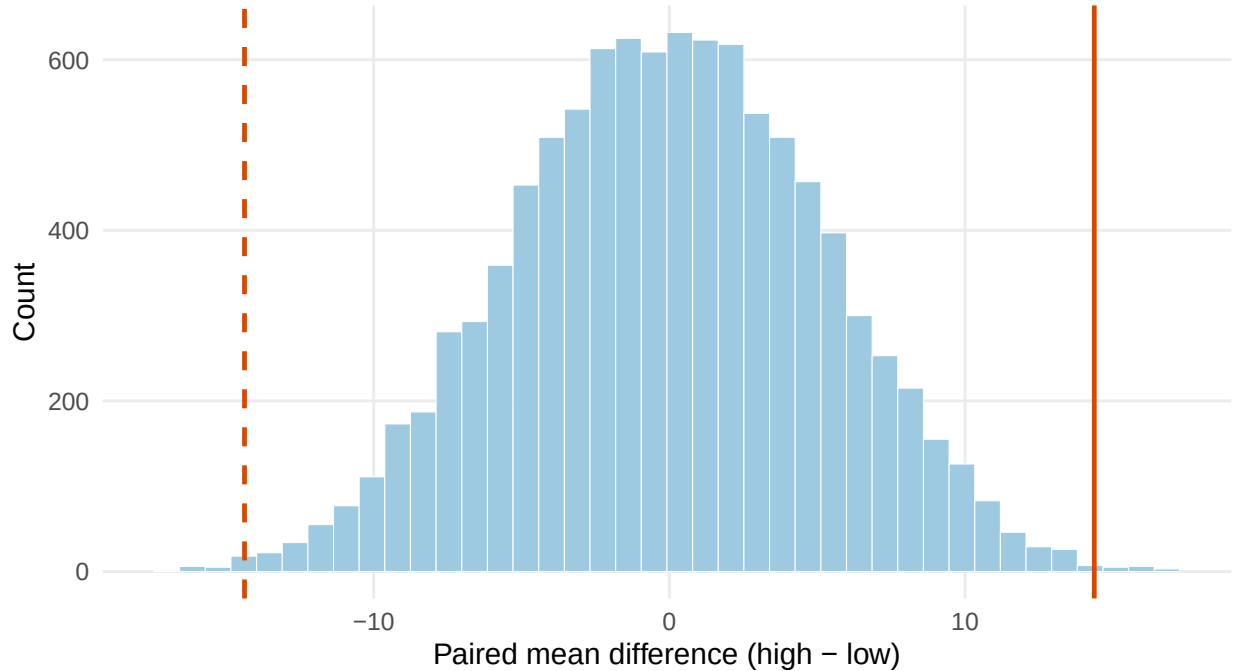
Table 35: Randomization inference summary

quantity	value
Simulations	10000.0000
Observed paired mean difference	14.3694
Two-sided RI p-value	0.0031
One-sided RI p-value (high > low)	0.0014

Figure

Randomization distribution under the sharp null

Solid orange = observed statistic; dashed orange = symmetric reference



The **stargazer** output for **R1** is included here as a compact regression restatement of the paired estimand. It is not the design-based test itself, but it makes the bridge from the paired-difference statistic to an intercept-only linear model explicit.

	Paired high-minus-low difference (seconds)
Average paired effect	14.369*** (4.692)
Observations	38

Note:

*p<0.1; **p<0.05; ***p<0.01

This table covers the regression member of the design-based inference family; R2 is the sign-flip randomization-inference procedure reported outside stargazer.

The intercept equals the observed mean paired treatment difference.

The practical conclusion is that the observed paired effect is rare under that randomization-based null. The RI p-value of 0.0031 closely reinforces the paired and fixed-effects results rather than contradicting them, which is exactly why this final check is valuable.

IX. Discussion and Conclusion

Across the primary specifications, the treatment effect is consistent in sign and similar in magnitude. The paired estimate, the participant fixed-effects model without period adjustment, the period-adjusted fixed-effects model, and the design-based randomization-inference check all point to the same practical conclusion: in this experiment, the high-tempo condition increased plank endurance by roughly fourteen seconds relative to the low-tempo condition.

The benchmark and precision sections clarify why the design mattered so much. A first-period-only parallel-arm analysis produced a much less precise estimate, a wider confidence interval, and a non-significant result. That contrast is not a contradiction. It is evidence that the randomized crossover design was the correct tool for this question because it absorbed the large baseline variance in plank performance that would otherwise overwhelm the treatment signal.

At the same time, the study has important limits. The protocol was remote and partially self-administered, so timing quality and adherence could not be supervised in person. The treatment was also narrow: one high-tempo instrumental EDM track and one low-tempo instrumental EDM track delivered under standardized instructions. For that reason, the experiment identifies the causal effect of this bundled treatment package more cleanly than it identifies the pure effect of tempo abstracted away from song identity, production style, or genre. A different pair of tracks, or a different genre entirely such as classical or heavy metal, might produce different results.

Generalizability should therefore be stated carefully. The findings speak most directly to short-duration isometric endurance in a remote plank task, not automatically to longer cardiovascular exercise such as running or cycling. What the experiment does show is that, in this setting, a carefully standardized high-tempo music condition improved performance relative to a carefully standardized low-tempo condition, and that the crossover design was essential for estimating that effect precisely.

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